Solving a sequence of eigenvalue problems using snapshot and row subsampling

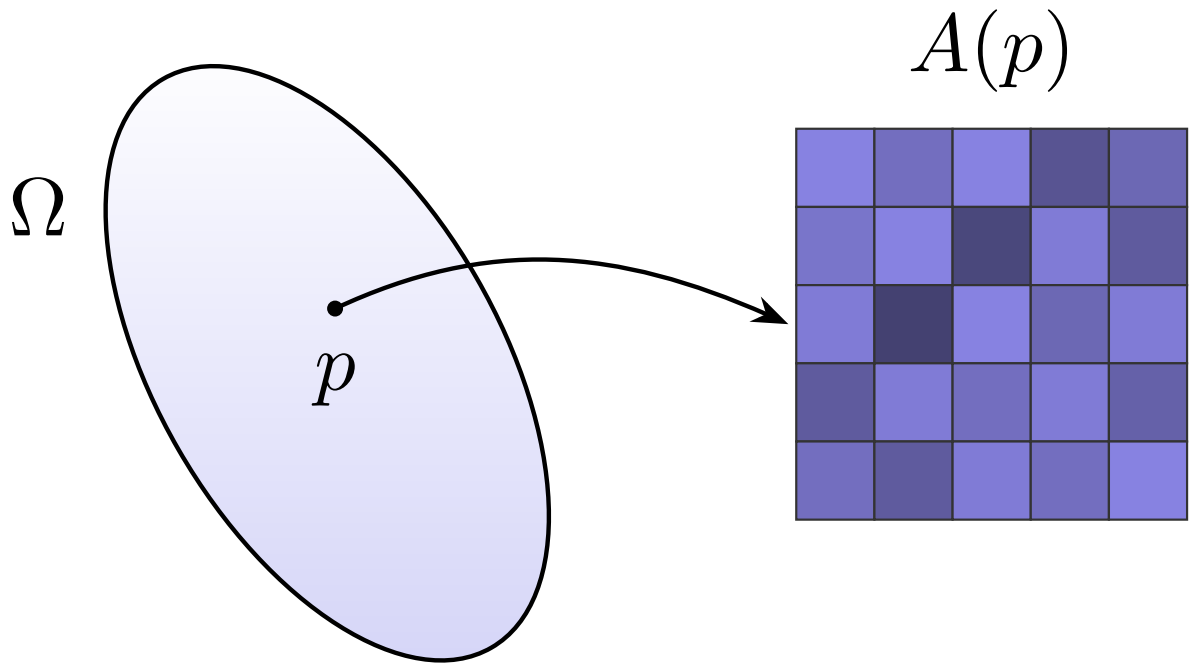
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joint work with Yuji Nakatsukasa

ENLARGED, Rutherford Appleton Laboratory
September 14, 2026



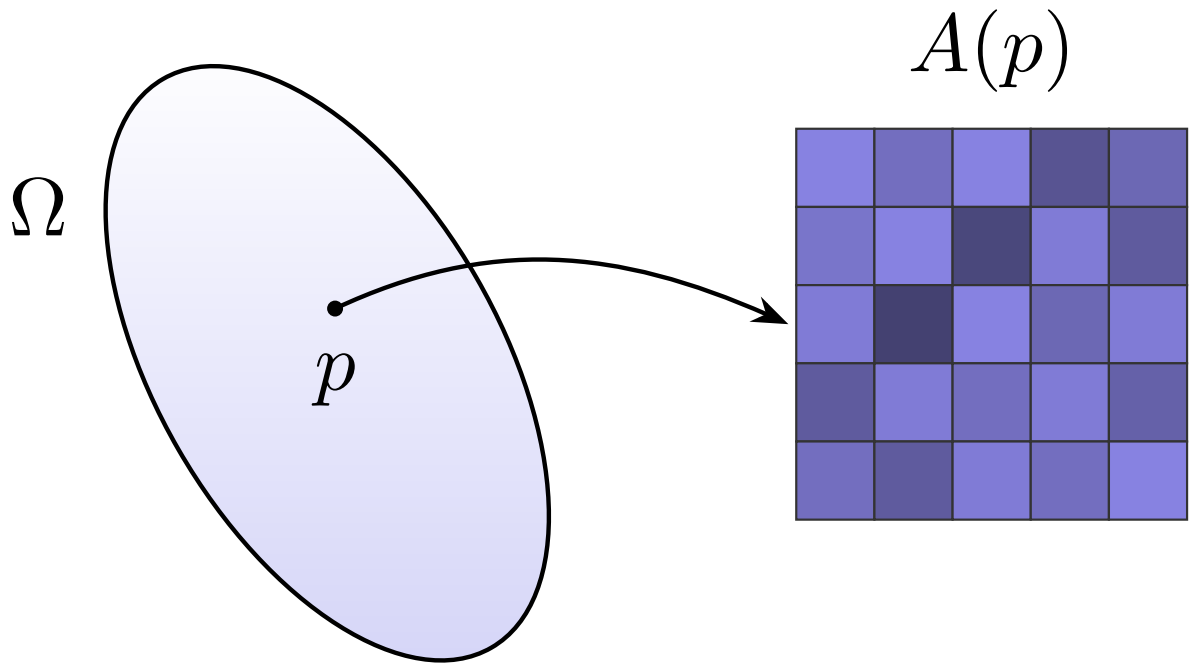
Suppose we have a function

$$\Omega \subset \mathbb{C} \mapsto \mathbb{C}^{n \times n}$$
$$p \mapsto A(p)$$



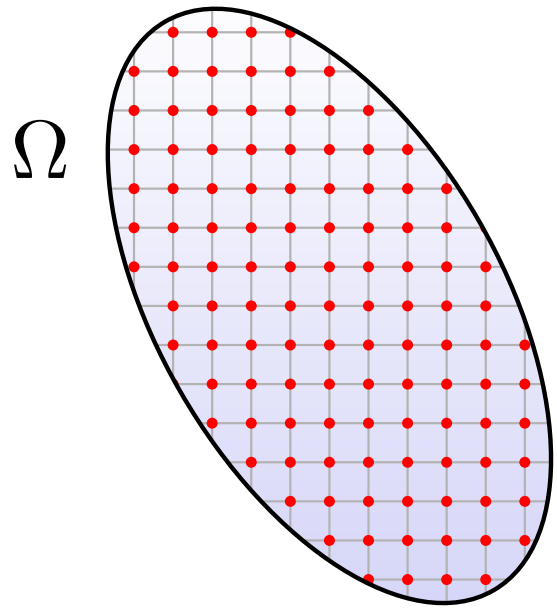
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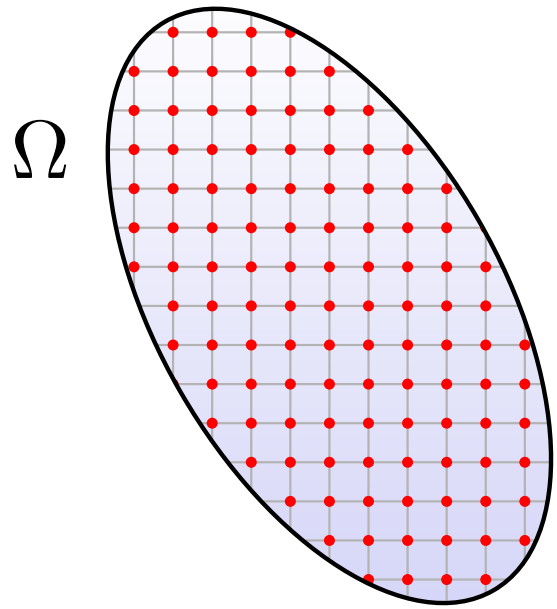


We want to compute $\lambda_{\min}(A(p))$ or $\lambda_{\max}(A(p))$ (and the corresponding eigenvector) for each value of p in the domain Ω.

Numerically, Ω is discretized with $N \gg 1$ points, and we want to solve an eigenvalue problem for N different matrices.



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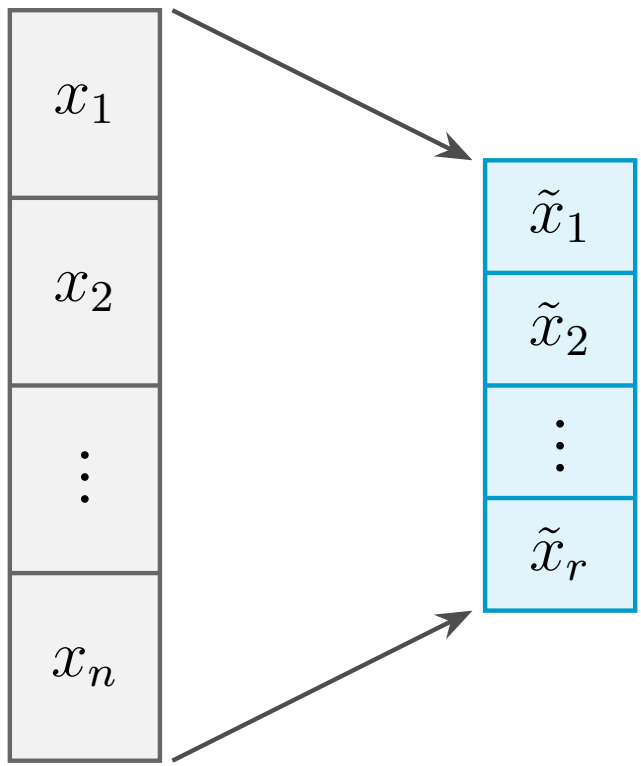


We must exploit the fact that the matrices are related to each other through the function A.

Sequences of eigenvalue problems appear in numerous applications:

- PDEs (Schmid and Henningson. 2001, §3);
- Computational chemistry (Le Bris 2005);
- Network science (Langville and Meyer 2012);
- Data analysis (Hastie, Tibshirani, and Friedman 2009);
- Kernel methods (Schölkopf and Smola 2002);
- ....

The main idea to solve this type of problem is the following: choose an orthonormal basis $Q \in \mathbb{C}^{n \times r}$, $r \ll n$, and search for the eigenvectors of $A(p)$ within $\text{span}(Q)$, reducing the search space dimension from n to r.



- How do we choose Q?
- Why does the chosen basis Q work?

} **snapshot**

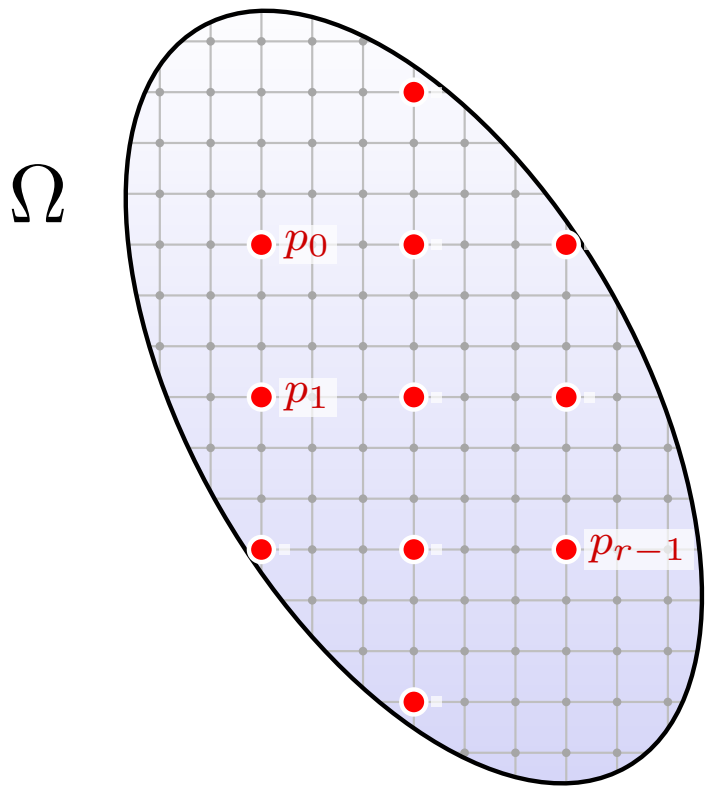
- How do we extract the eigenvector of $A(p)$ from Q for an unanalyzed parameter p?

} **row subsampling**

How do we choose Q?

A classical choice for Q is:

- Choose r parameter values $p_0, \ldots, p_{r-1}$;
- Compute the eigenvectors of $A(p_i)$, $i = 0, \ldots, r-1$, denoted by $\boldsymbol{x}(p_i)$.
- Choose Q as an orthonormal basis of $\text{span}\{\boldsymbol{x}(p_0), \ldots, \boldsymbol{x}(p_{r-1})\}$.



Why does the chosen basis Q work?

Assume the domain Ω is $[-1, 1]$ and that $A(p)$ can be extended to an analytic function on the Bernstein ellipse E_ρ, $\rho > 1$.

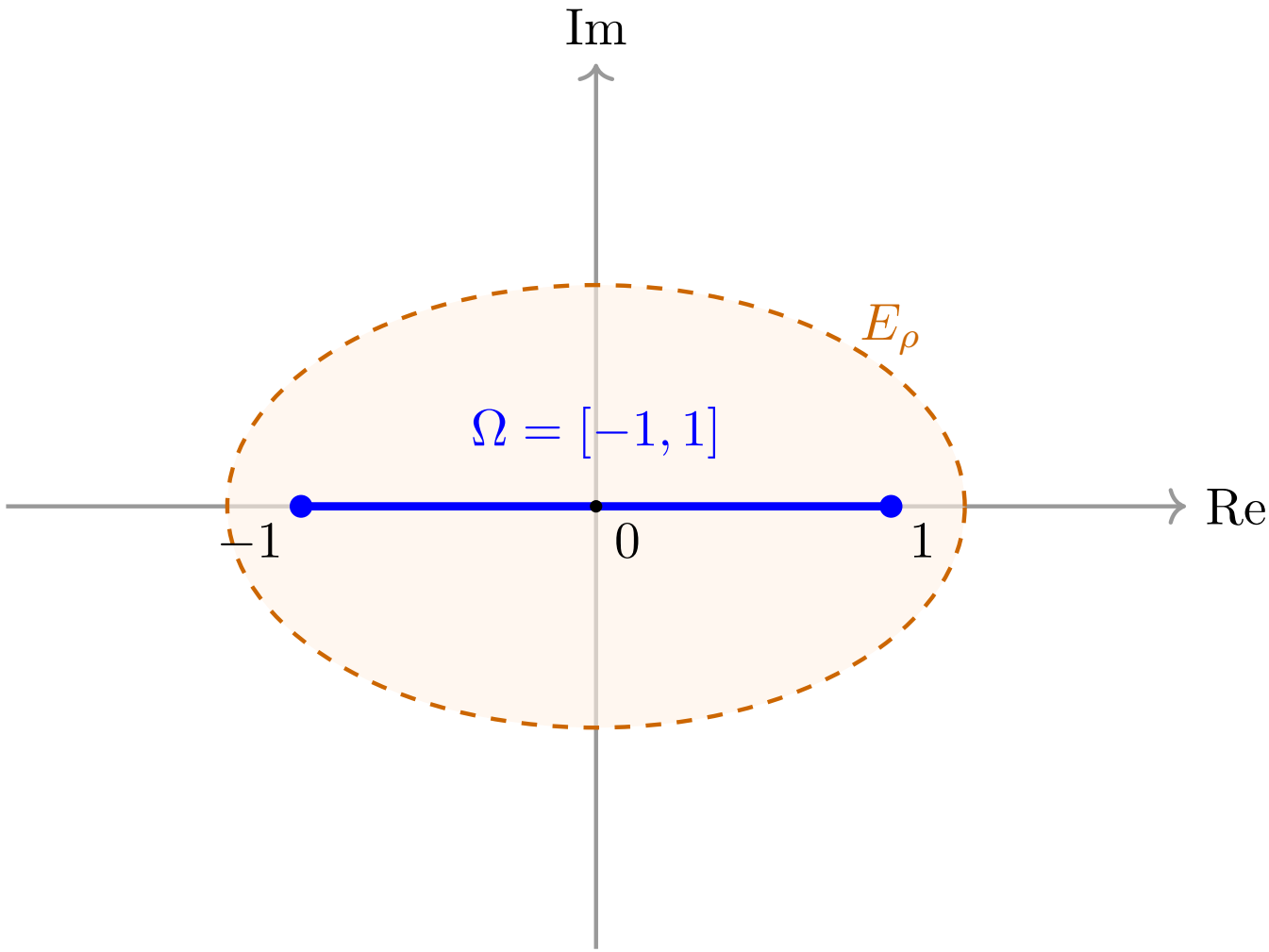
- If the parameters $p_0, \ldots, p_{r-1}$ are equidistant, then

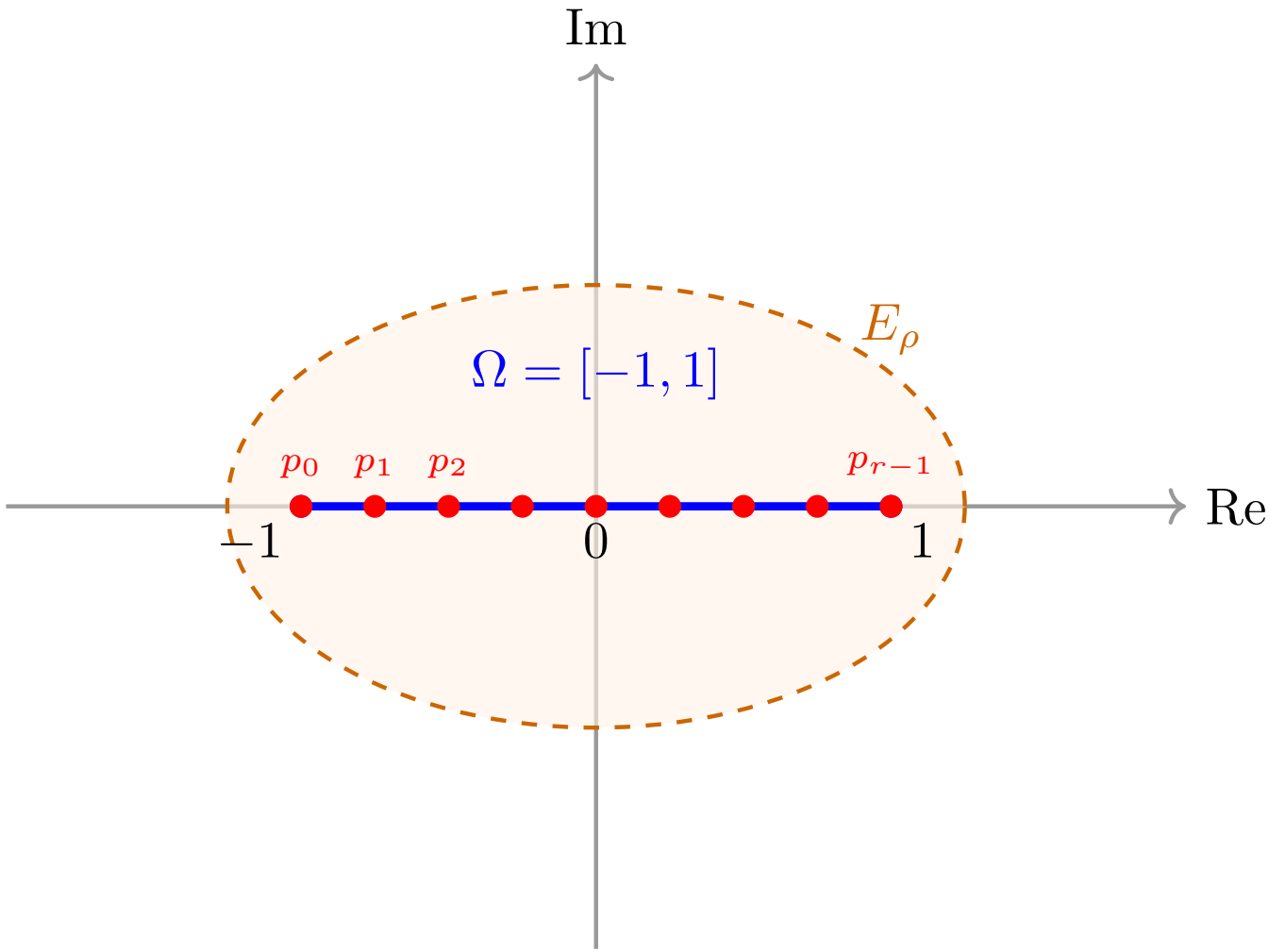
$$\|(I - QQ^*)\boldsymbol{x}(p)\| \lesssim \rho^{-\sqrt{r}}$$

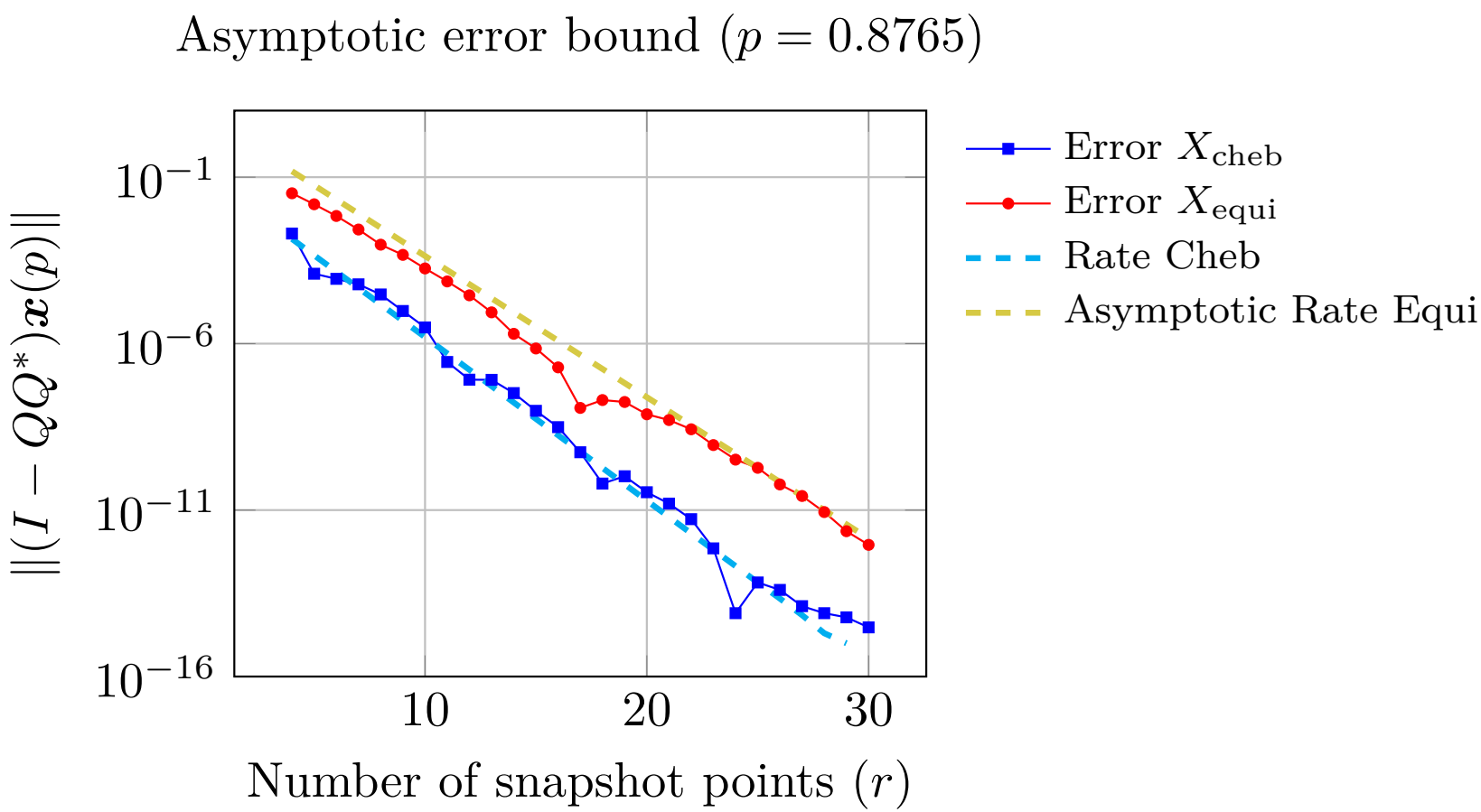
- If the parameters $p_0, \ldots, p_{r-1}$ are Chebyshev points, then

$$\|(I - QQ^*)\boldsymbol{x}(p)\| \lesssim \rho^{-r}$$

- For any distribution of parameters $p_0, \ldots, p_{r-1}$ such that the discrete measure converges to a measure μ in the weak-star topology, asymptotically in r, the quantity $\|(I - QQ^*)\boldsymbol{x}(p)\|$ decays exponentially (provided that ρ is sufficiently large).


Im
Re
E_ρ
$\Omega = [-1, 1]$
-1
0
1


Im
Re
E_ρ
$\Omega = [-1, 1]$
p_0
p_1
p_2
p_{r-1}
-1
0
1


Asymptotic error bound ($p = 0.8765$)
$\|(I - QQ^*)\boldsymbol{x}(p)\|$
10^{-1}
10^{-6}
10^{-11}
10^{-16}
10
20
30
Number of snapshot points (r)
Error X_{cheb}
Error X_{equi}
Rate Cheb
Asymptotic Rate Equi

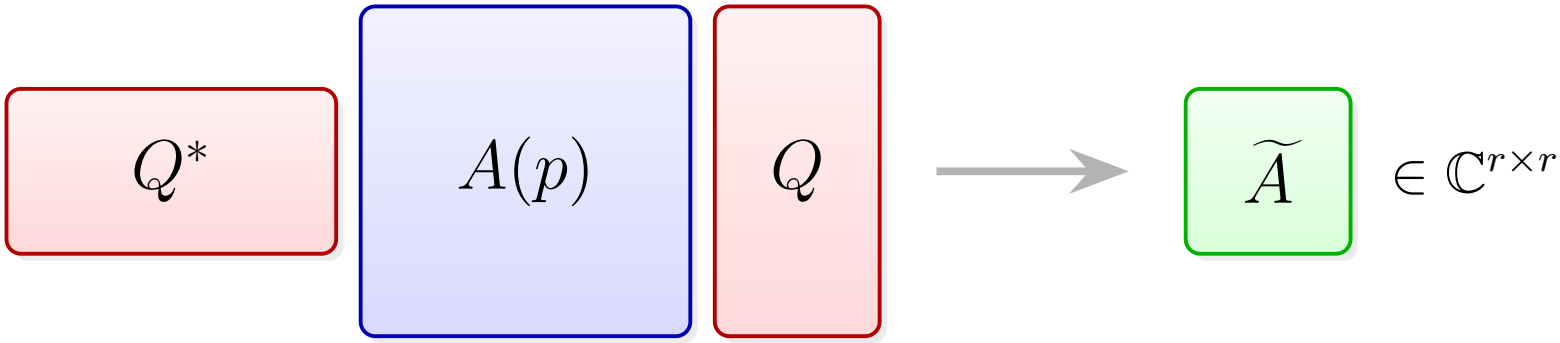
How do we extract the eigenvector from Q?

One of the most common methods is the **Rayleigh-Ritz** method: project $A(p)$ as $Q^*A(p)Q \in \mathbb{C}^{r\times r}$, find an eigenvector $\boldsymbol{y}(p)$ of this matrix

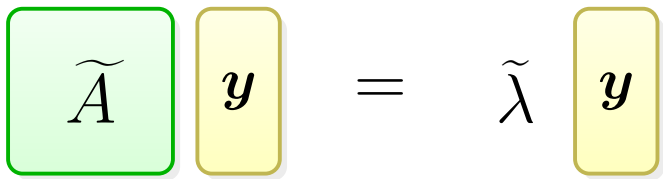
$$Q^*A(p)Q\boldsymbol{y}(p) = \tilde{\lambda}(p)\boldsymbol{y}(p)$$

and return the pair $(\tilde{\lambda}(p), Q\boldsymbol{y}(p))$.

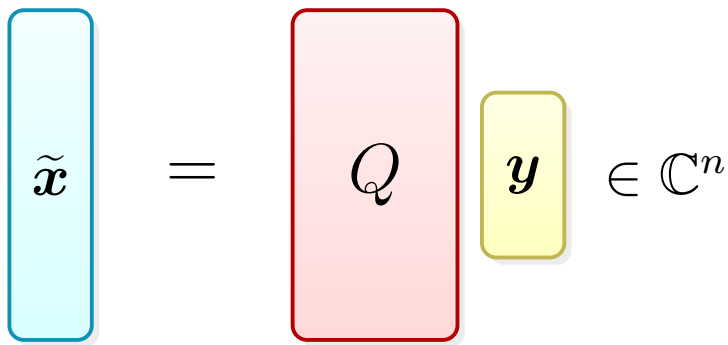
1. Projection



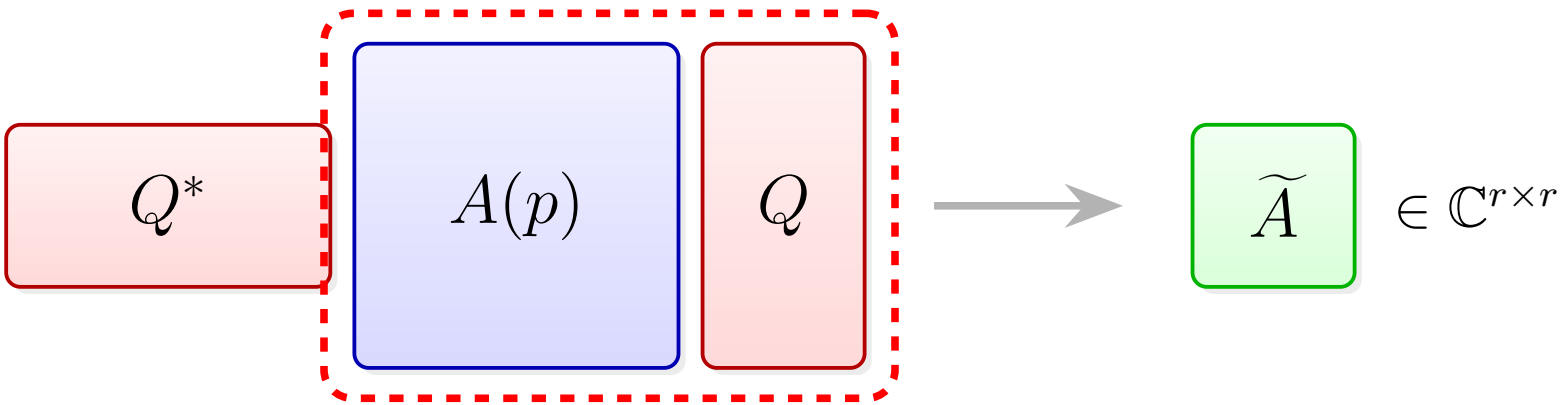
2. Solution (In the reduced space)



3. Eigenvector reconstruction



The weak point



Note that the matrix $Q^* A(p) Q$ is the solution of

$$\text{argmin}_{M \in \mathbb{C}^{r \times r}} \|A(p)Q - QM\|_F.$$

This is a tall and skinny least-squares problem, one of the problems where randomized linear algebra has had a major impact.

Instead of solving

$$\arg\min_{M} \|A(p)Q - QM\|_F$$

we solve

$$\arg\min_{M} \|S(A(p)Q - QM)\|_F$$

where $S \in \mathbb{R}^{s\times n}$, $s \geq r$, is a row-selection matrix.

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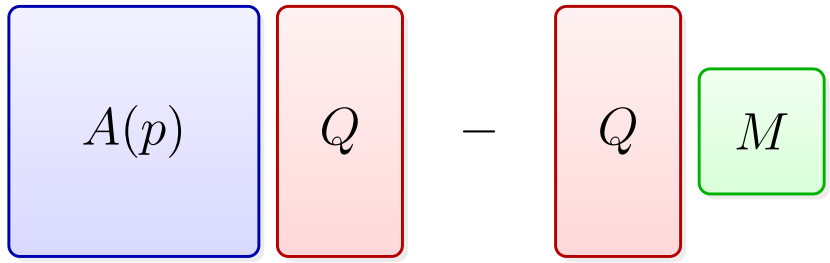
we solve

$$\arg\min_{M} \|S(A(p)Q - QM)\|_F$$

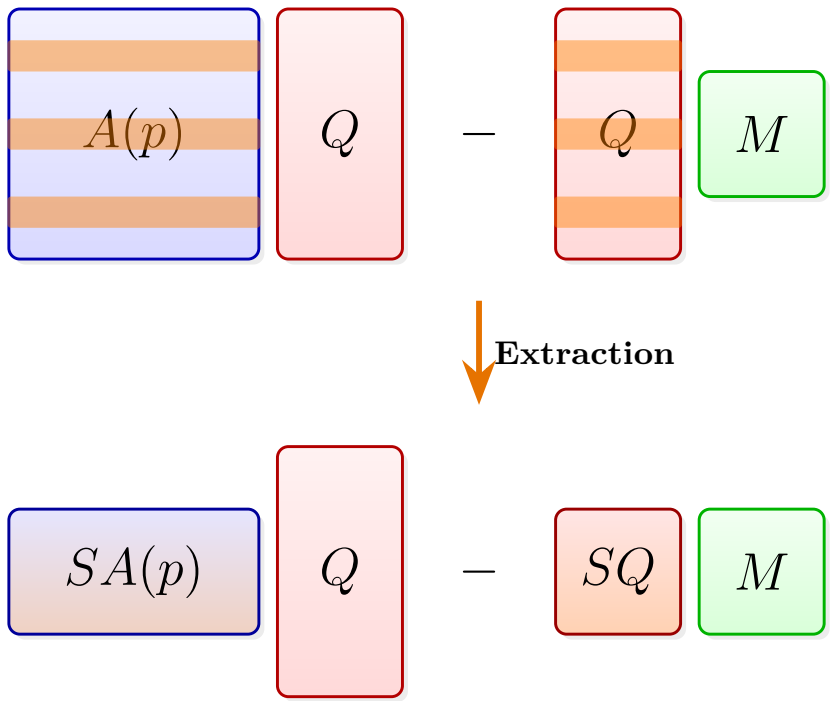
where $S \in \mathbb{R}^{s \times n}$, $s \geq r$, is a row-selection matrix.

- Theoretical guarantees on the residual;
- Only a few rows of $A(p)$ are required;
- Very fast solution.

1. Original Problem: $\arg\min_M \|A(p)Q - QM\|_F$



2. Row selection via S



Algorithm 1:

Offline Phase

- Compute Q by solving the full eigenvalue problem for $A(p_0), \ldots, A(p_{r-1})$
- Select rows from Q

Online Phase (for each new p)

- Solve the least-squares optimization: $M(p) = \arg\min_{M} \|SA(p)Q - SQM\|_F$
- Compute the desired eigenvector of the reduced matrix $M(p)$

Extension to SVD

A possible generalization is to compute two subspaces Q_ℓ and Q_r and compute the singular vectors and values of $Q_\ell^* A(p) Q_r$. Following the previous reasoning, this is accelerated by solving the least-squares problem

$$\|SA(p)Q_r - SQ_\ell M\|_F.$$

Another option is to set $Q_r = I$ (equivalent to Rayleigh-Ritz on AA^*) and solve

$$\|SA(p) - SQ_l M\|_F.$$

Example 1: Exponential differential operator

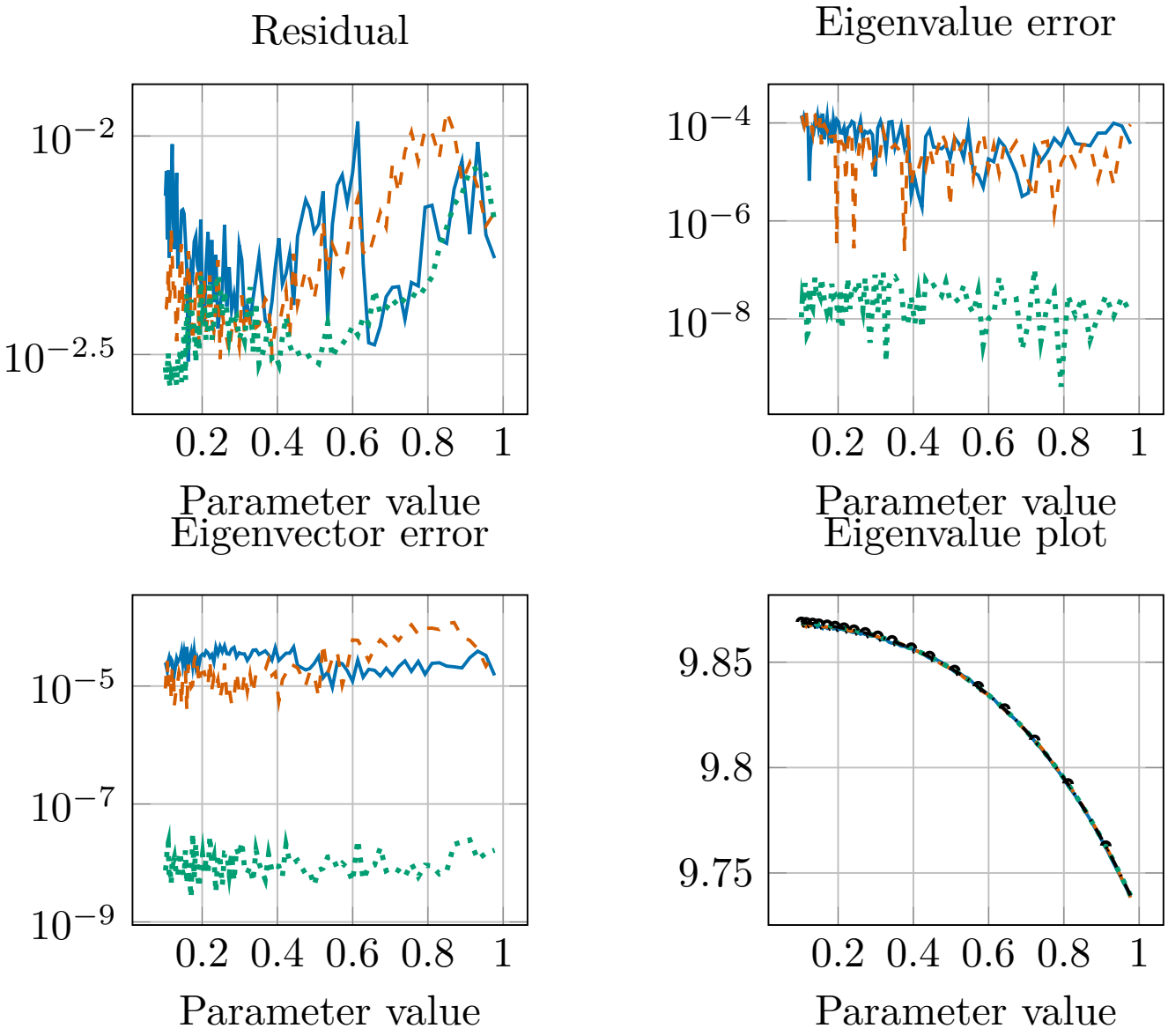
In the first example, we compute the smallest eigenvalue of the 3-point stencil finite-difference discretization of the differential operator

$$\frac{d}{dx}\left(e^{px}\frac{du}{dx}\right) \qquad x \in \Omega = [-10, 10],$$

with zero boundary conditions.

- The parameter space $p \in [0.1, 1]$ is discretized using 3000 log-spaced points.
- Domain Ω is discretized such that the resulting matrix size is $n = 2 \times 10^6$.
- We use 10 log-spaced snapshot points in the parameter space.

Example 1


Residual
10^{-2}
$10^{-2.5}$
0.2 0.4 0.6 0.8 1
Parameter value
Eigenvalue error
10^{-4}
10^{-6}
10^{-8}
0.2 0.4 0.6 0.8 1
Parameter value
Eigenvector error
10^{-5}
10^{-7}
10^{-9}
0.2 0.4 0.6 0.8 1
Parameter value
Eigenvalue plot
9.85
9.8
9.75
0.2 0.4 0.6 0.8 1
Parameter value
Leverage
LU
POD
Exact

Example 1

Method	Offline(s)	Online(s)	Total(s)	Ratio Off.	Ratio On.	Ratio Tot.
LU	9.282	64.555	73.837	0.999	**0.047**	0.053
Leverage	9.224	71.113	80.337	0.993	**0.052**	0.058
POD	9.288	1,378.037	1,387.325	1.000	**1.000**	1.000

Example 2: Differential operator with multiple eigenvalues

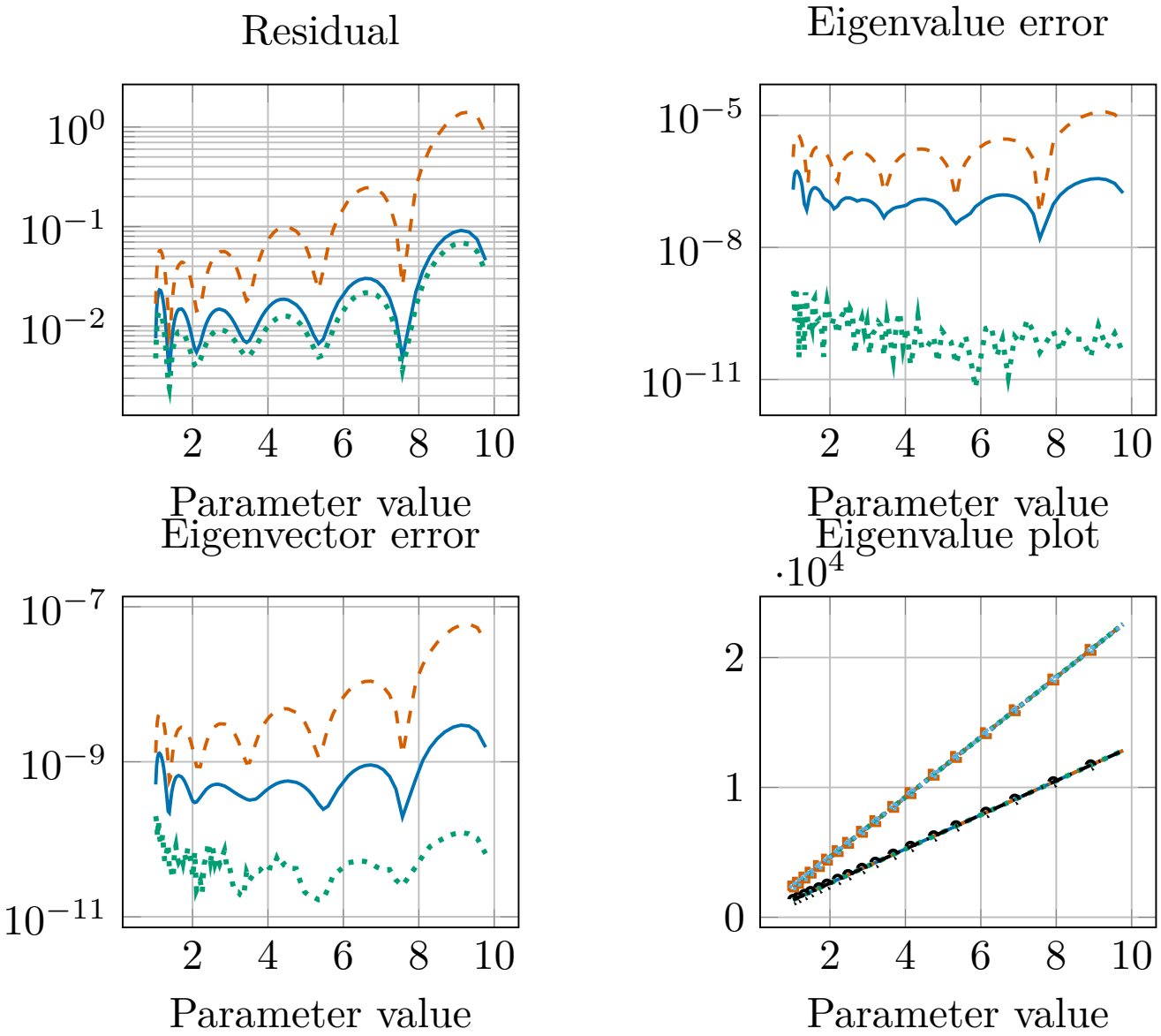
We compute the smallest 3 eigenvalues of the 5-point stencil finite-difference discretization of the operator

$$-p\Delta\mathbf{u} - \nabla(\nabla \cdot \mathbf{u}) \qquad \Omega = [-1, 1]^2,$$

with zero boundary conditions.

- Dimension of the matrix: $n = 250000$.
- The second smallest eigenvalue has multiplicity 2 due to system symmetry.
- Parameter space: p log-spaced between 10^{-4} and 10^{-3}.
- We use 10 log-spaced snapshot points.

Example 2


Residual
10^0
10^{-1}
10^{-2}
2 4 6 8 10
Parameter value
Eigenvalue error
10^{-5}
10^{-8}
10^{-11}
2 4 6 8 10
Parameter value
Eigenvector error
10^{-7}
10^{-9}
10^{-11}
2 4 6 8 10
Parameter value
Eigenvalue plot
$\cdot 10^4$
2
1
0
2 4 6 8 10
Parameter value
Leverage
LU
POD
Exact

Example 2

Method	Offline(s)	Online(s)	Total(s)	Ratio Off.	Ratio On.	Ratio Tot.
LU	249.480	11.512	260.992	0.977	**0.025**	0.364
Leverage	254.718	26.565	281.283	0.997	**0.058**	0.392
POD	255.457	461.676	717.133	1.000	**1.000**	1.000

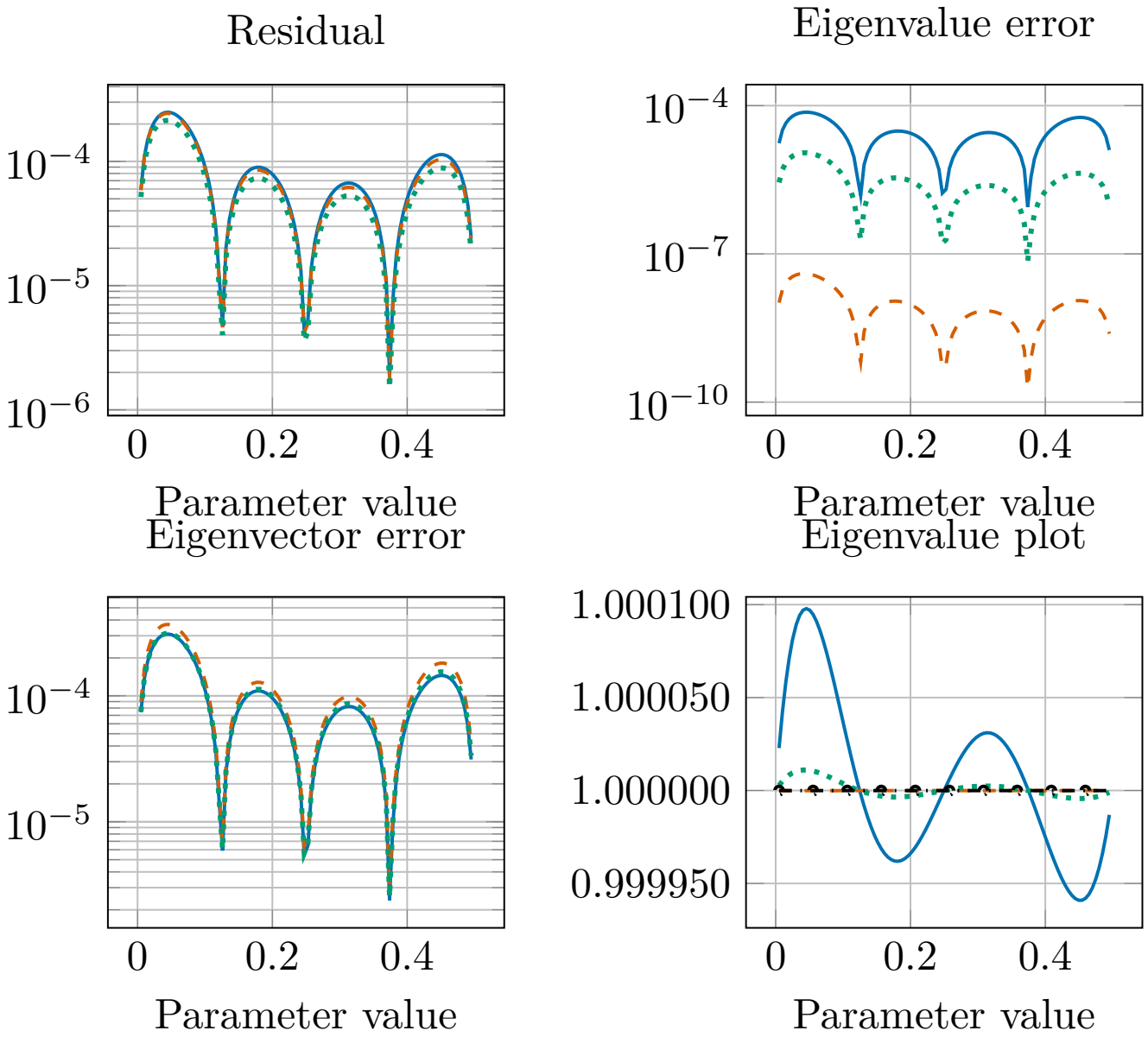
Example 3: PageRank

We compute the leading eigenpair of a degree-biased parametric PageRank problem on the `web-NotreDame` dataset ($n = 325729$):

$$A(p) = dW(p) + \frac{1-d}{n}\mathbf{1}\mathbf{1}^T.$$

- Base transition matrix: $W(p)$ obtained by column-normalizing $M_{ij}(1 + \text{in_degree}_i)^p$.
- Fixed damping factor $d = 0.9$.
- Parameter space: 1000 equispaced points in $p \in [0, 0.5]$.
- We use 5 equispaced snapshot points across the parameter domain.

Example 3


Residual
10^{-4}
10^{-5}
10^{-6}
0
0.2
0.4
Parameter value
Eigenvalue error
10^{-4}
10^{-7}
10^{-10}
0
0.2
0.4
Parameter value
Eigenvector error
10^{-4}
10^{-5}
0
0.2
0.4
Parameter value
Eigenvalue plot
1.000100
1.000050
1.000000
0.999950
0
0.2
0.4
Parameter value
Leverage
LU
POD
Exact

Example 3

Method	Offline(s)	Online(s)	Total(s)	Ratio Off.	Ratio On.	Ratio Tot.
LU	12.157	11.741	23.898	1.030	**0.4532**	0.634
Leverage	12.761	12.065	24.825	1.081	**0.4657**	0.658
POD	11.805	25.906	37.711	1.000	**1.0000**	1.000

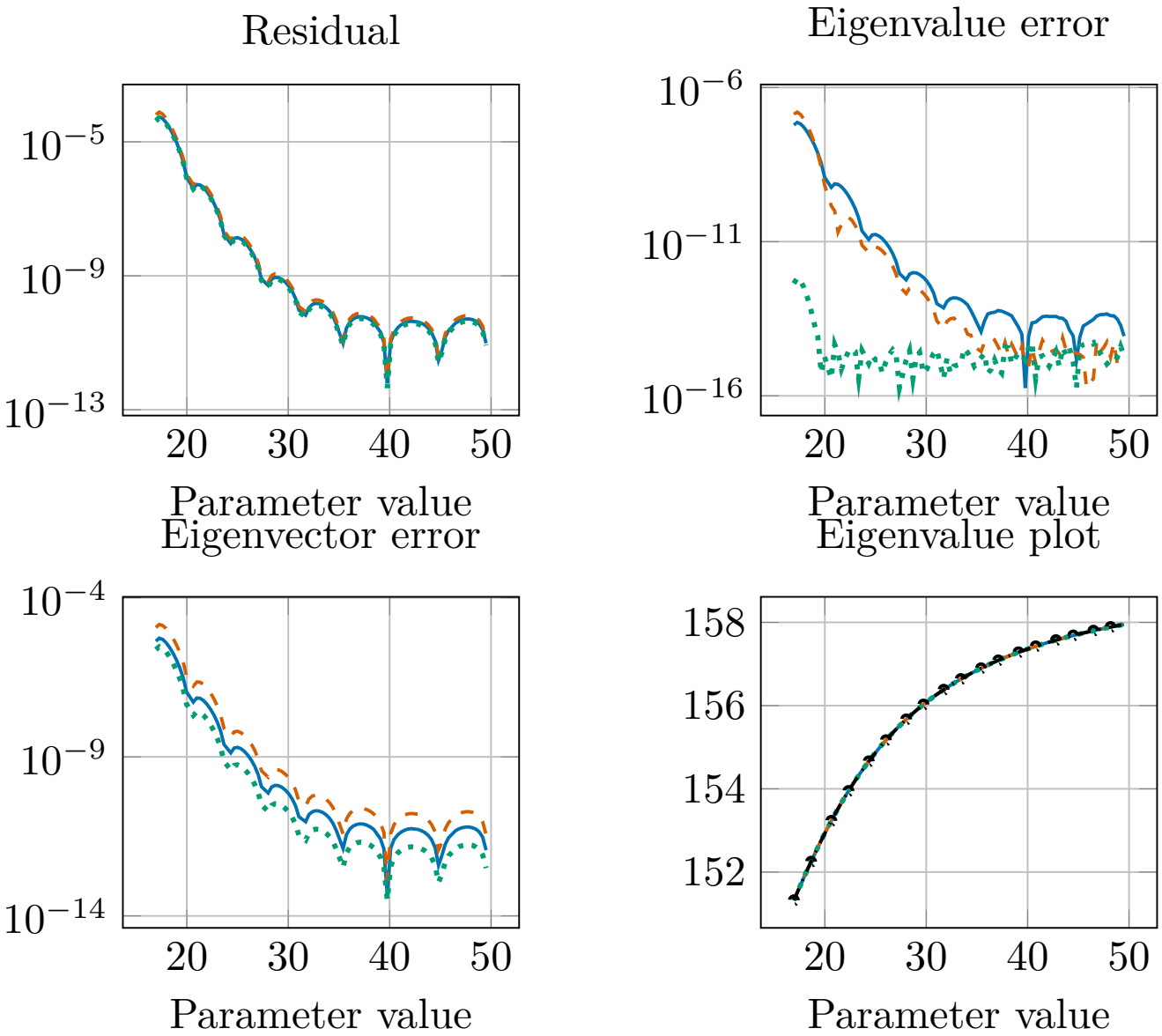
Example 4: Gaussian Kernel

We apply the algorithm to the Gaussian kernel matrix produced from the MNIST dataset ($n = 60000$):

$$A_{ij}(p) = \exp\left(-\frac{\|x_i - x_j\|^2}{2p^2}\right).$$

- Entrywise exponential based on pairwise squared Euclidean distances.
- Parameter space: 1000 equispaced points in $[0.5p_*, 1.5p_*]$.
- $p_* = \frac{\log(2)}{\text{median}}$ where median is the median distance between distinct points.
- We use 10 equispaced snapshot points.

Example 4


Residual
10^{-5}
10^{-9}
10^{-13}
20
30
40
50
Parameter value
Eigenvalue error
10^{-6}
10^{-11}
10^{-16}
20
30
40
50
Parameter value
Eigenvector error
10^{-4}
10^{-9}
10^{-14}
20
30
40
50
Parameter value
Eigenvalue plot
158
156
154
152
20
30
40
50
Parameter value
Leverage
LU
POD
Exact

Example 4

Method	Offline(s)	Online(s)	Total(s)	Ratio Off.	Ratio On.	Ratio Tot.
LU	13.033	1.882	14.915	1.437	**0.004**	0.030
Leverage	13.329	1.993	15.322	1.470	**0.004**	0.031
POD	9.070	484.148	493.218	1.000	**1.000**	1.000

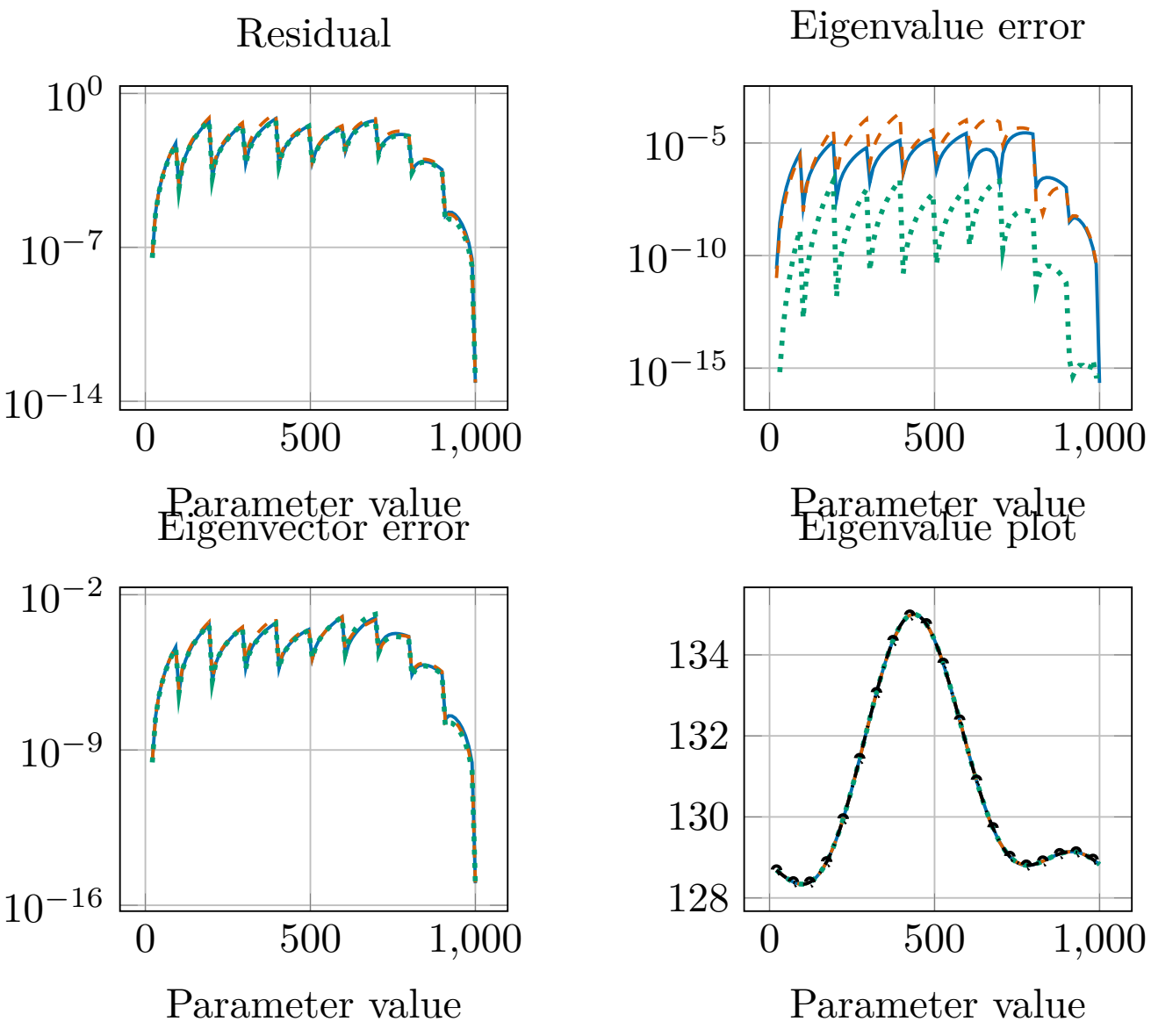
Example 5: Graph Eigenvector Centrality

We compute the dominant eigenpair to evaluate eigenvector centrality in the `coAuthors-DBLP` network ($n = 299067$):

$$A(p) = D(p)A_0D(p).$$

- A_0: adjacency matrix of the network graph.
- $D(p)_{i,i} = 1 + 0.5\sin(2\pi p/1000 + \Phi_i)$, with $\Phi_i \sim \mathcal{U}[0, 2\pi]$.
- Parameter space: 1000 integer points uniformly spaced in $[1, 1000]$.
- We use 20 snapshot points concentrated at the start with updates every 100 steps.

Example 5: Update freq. = 100



Example 5: Update freq. = 100

Method	Offline(s)	Online(s)	Total(s)	Ratio Off.	Ratio On.	Ratio Tot.
LU	7.214	10.133	17.346	1.079	**0.176**	0.270
Leverage	6.800	10.394	17.194	1.017	**0.181**	0.268
POD	6.684	57.575	64.259	1.000	**1.000**	1.000

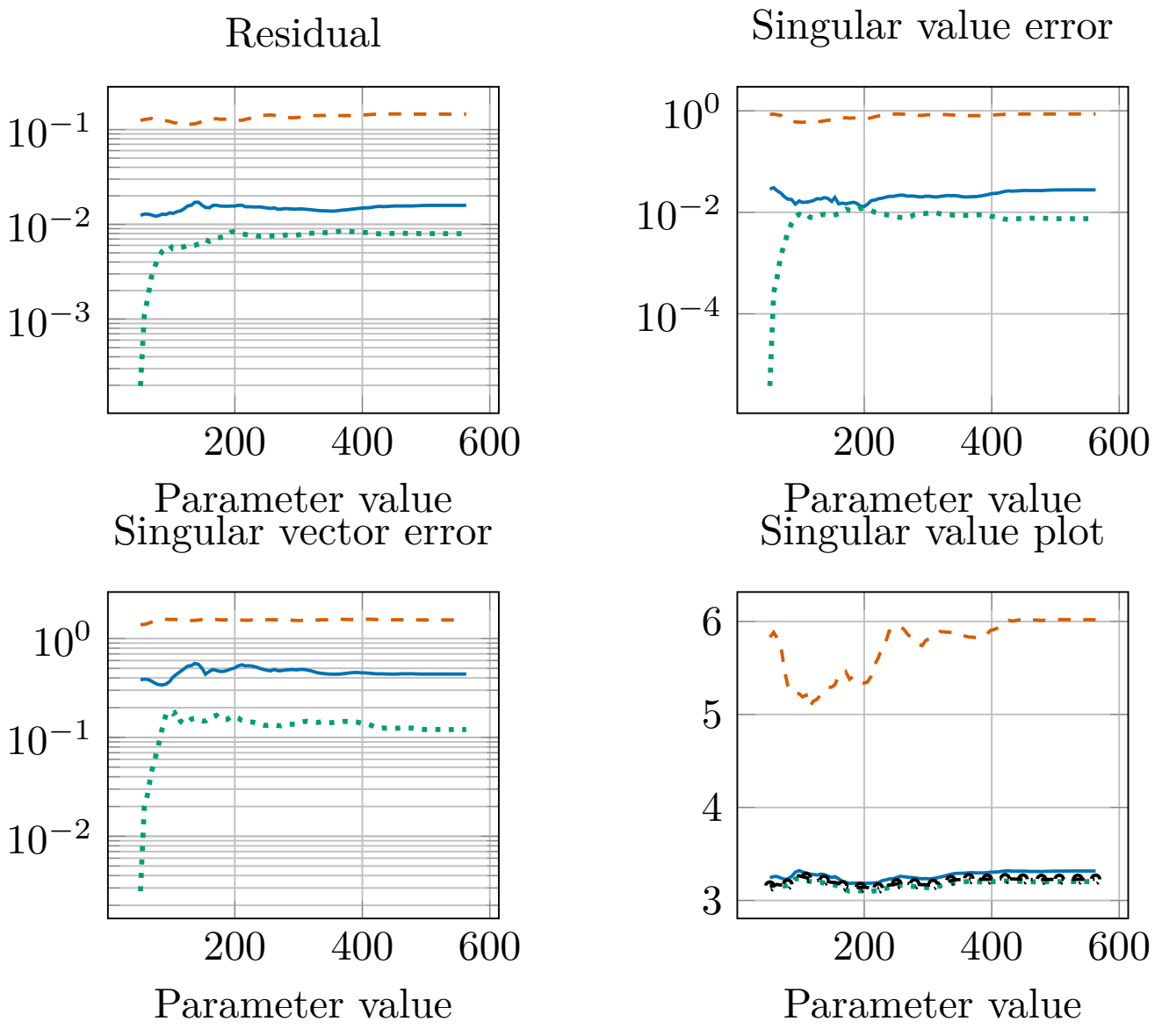
Example 6: Transformer

We track the spectral evolution of the attention matrix during fine-tuning of *ModernBERT-base* on the LexGLUE SCOTUS dataset (document length 5000 tokens):

$$\text{softmax}\left(\frac{XW_Q(XW_K)^T}{\sqrt{64}}\right).$$

- Focus on 3rd attention head of 21st layer ($W_Q, W_K \in \mathbb{R}^{768\times 64}$).
- Tracking leading singular value and vector over fine-tuning steps.
- Parameter space: 563 sequential update steps ($p \in \{1, \ldots, 563\}$) during epoch 4.
- We use 50 snapshot points $(1, \ldots, 50)$ concentrated at the beginning.

Example 6


Residual
10^{-1}
10^{-2}
10^{-3}
200
400
600
Parameter value
Singular value error
10^{0}
10^{-2}
10^{-4}
200
400
600
Parameter value
Singular vector error
10^{0}
10^{-1}
10^{-2}
200
400
600
Parameter value
Singular value plot
6
5
4
3
200
400
600
Parameter value
Leverage
LU
POD
Exact

Example 6

Method	Offline(s)	Online(s)	Total(s)	Ratio Off.	Ratio On.	Ratio Tot.
LU	13.311	2.893	16.205	0.831	**0.032**	0.151
Leverage	13.040	4.711	17.751	0.815	**0.051**	0.165
POD	16.010	91.644	107.654	1.000	**1.000**	1.000

Code available at
https://github.com/fhrobat/SubApSnapEig

Preprint soon available!

Thank You!

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